Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China

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Abstract

We compare spatial wet-bulb temperature contrast between upper-quartile and middle-half days of the monthly regional mean at 121 reanalysis sites in eastern China. We developed the statistic and field-selection rules using the 2015 and 2022 summers, then applied them unchanged to the remaining 33 summers in 1991–2025. The mean contrast across five graph scales was −7.28%; 28 of 33 summer contrasts were negative. An exploratory 99,999-draw product cyclic-shift analysis gave $p = 0.00001$ under the conditional invariance null. Contraction strengthened from −2.86% at 126 km to −13.27% at 2,013 km. An exact Laplacian decomposition showed that high-day anomalies opposed the monthly climatological pattern at broad scales, whereas transient-anomaly energy changed little. In ten summers excluded from method development, surface stations agreed in direction at broad scales (−17.34%; nine of ten contrasts negative), whereas uncertainty at 126 km included zero. Applying the same protocol to 1950–1990 gave −11.04%, with 40 of 41 summer contrasts negative and the same strengthening with bandwidth. High regional mean wet-bulb temperature was therefore associated with weaker broad geographic contrast, an aspect that a regional mean cannot describe.

Keywords: ERA5-Land; graph Laplacian; humid heat; randomisation inference; semivariance; spatial statistics

1 Introduction

Wet-bulb temperature (WBT) is a widely used measure of humid heat stress that combines air temperature and humidity (Sherwood and Huber, 2010; Raymond et al., 2020). Two days can have identical regional means yet sharply different spatial fields: one may contain strong north–south or local contrasts, while the other is comparatively coherent. We therefore focus on the spatial contrast of the WBT field itself.

Recent work has documented increasingly concentrated humid-heat extremes in standard climate regions (Speizer et al., 2022), sharp local wet-bulb extremes associated with wet soils (Chagnaud et al., 2025), and reduced intra-urban heat-stress variability during some heat waves (Shreevastava et al., 2023). Spatial-extremes models such as Healy et al. (2025) estimate nonstationary temperature tails and the area exceeding critical thresholds. Our within-month comparison instead examines how pairwise WBT differences change with graph scale. A regional mean discards spatial arrangement, while ordinary cross-site variance cannot distinguish neighbouring contrasts, mesoscale structure and a domain-wide gradient. Empirical variograms represent distance-dependent variation, but comparing regimes requires a common summary across distance scales. Smooth weighting provides that summary and permits an exact spatial allocation of the regional contrast.

We study whether high regional mean wet-bulb temperature in eastern China is accompanied by a contraction of spatial differences. The question is associational, which makes inference delicate. A negative contrast is compatible with broad atmospheric forcing, but topography (Pepin et al., 2015), soil moisture (Seneviratne et al., 2010), circulation (Horton et al., 2015), and measurement support can produce or mask similar patterns.

If event labels depend on spatial dispersion as well as the regional mean, the outcome influences which days are selected, and a difference can be created by construction (Simmons et al., 2011). If the most favourable distance is chosen after inspecting the data, scale-specific evidence is likewise overstated (White, 2000). We therefore define events from the regional mean wet-bulb temperature alone and fix the spatial scales before evaluating any dispersion outcome.

We combine established graph Dirichlet energies into a scale-indexed regime contrast for a fixed observational record. Under second-order stationarity, each energy is a smoothly weighted semivariance (Cressie, 1993; Shuman et al., 2013; Dong et al., 2016). The construction averages record-specific relative contrasts over five log-spaced graphs, preserves the daily multiscale profile in product cyclic shifts, and decomposes the regional effect exactly over nodes and low-dimensional spatial components. This combination reveals which geographic scales and structures account for a change in WBT contrast. Ordinary variance discards distance, binned semivariances impose hard boundaries, and Moran's I and Geary's C measure standardised dependence rather than the amplitude of spatial differences.

We developed the primary estimand, event labels, bandwidths and field-selection rules using the 2015 and 2022 summers, then applied them unchanged to the other 33 summers (White, 2000). The resulting mean describes that finite record. Product cyclic shifts assess the fixed multiscale statistic under an explicit conditional invariance null, while equal edge splitting and Laplacian-orthogonal projections distinguish attenuation of persistent geographic structure from smoothing of transient anomalies.

2 Multiscale graph regime contrasts

2.1 Scale-indexed graph dispersion and regime contrast

Let y_{it} denote WBT at location i in the selected field for day t, and let $\bar{y}_t = n^{-1}\sum_i y_{it}$. The benchmark is the fixed-site mean squared spatial deviation,

$$V_t = \frac{1}{n}\sum_{i=1}^{n}(y_{it} - \bar{y}_t)^2, \qquad n = 121. \tag{1}$$

To retain spatial arrangement, we use an equirectangular kilometre projection centred at the mean site latitude: longitude is multiplied by $111.32\cos(\bar{\phi})$ and latitude by 110.57. For bandwidth h, set $w_{ij,h} = \exp\{-d_{ij}^2/(2h^2)\}$ for $i \neq j$ and $w_{ii,h} = 0$. Let $\mathbf{W}_h = (w_{ij,h})$, $\mathbf{L}_h = \mathrm{diag}(\mathbf{W}_h\mathbf{1}) - \mathbf{W}_h$, and $S_h = \sum_{i<j} w_{ij,h}$. We define graph dispersion as

$$Q_h(\mathbf{y}) = \frac{\mathbf{y}^\top \mathbf{L}_h \mathbf{y}}{2S_h} = \frac{\sum_{i<j} w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} w_{ij,h}}. \tag{2}$$

Graph dispersion is therefore an edge-weighted average of pairwise squared differences. Dividing by S_h makes the measure comparable across graphs with different total edge weights. Smaller values indicate greater graph-scale coherence. Graph dispersion measures spatial contrast rather than threshold-exceedance area. It has units $°\mathrm{C}^2$; $D_h^{\mathrm{RMS}} = (2Q_h)^{1/2}$ is the weighted root mean squared pairwise WBT difference in °C.

Equation (2) connects three familiar descriptions of spatial variation. It is the graph Dirichlet energy used to measure graph signal roughness (Shuman et al., 2013; Dong et al., 2016). If the field is second-order stationary with semivariogram $\gamma(d_{ij}) = \frac{1}{2}\,\mathrm{E}(Y_i - Y_j)^2$, then

$$\mathrm{E}\{Q_h(\mathbf{Y})\} = \frac{\sum_{i<j} w_{ij,h}\gamma(d_{ij})}{S_h}, \tag{3}$$

so its expected profile is a kernel average of the semivariogram. This interpretation requires second-order stationarity; the finite-field statistic does not. In the nonstationary WBT fields studied here, Q_h is interpreted only as graph-scale dispersion. On a complete graph with equal off-diagonal weights, $Q_h = nV/(n-1)$. Ordinary spatial variance is therefore the complete-graph endpoint of the same family, up to its degrees-of-freedom factor.

We fixed five bandwidths during method development: $h/m \in \{0.125, 0.25, 0.5, 1, 2\}$, where m is the median pairwise distance. They correspond to 126, 252, 503, 1,006 and 2,013 km. Their weighted mean pair distances are 200, 319, 549, 826 and 992 km, and their effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. Thus 126 km denotes a Gaussian bandwidth with no hard distance cut-off; it is local only relative to the sampled support, where the median nearest-neighbour distance is 150 km. For edge weights w_e, the reported Kish quantity is $N_{\text{eff}} = (\sum_e w_e)^2 / \sum_e w_e^2$.

Classifying days separately within each month-year record $r = (y, m)$ prevents the seasonal cycle or an interannual mean shift from entering the event definition. Let $q_{25,r}$ and $q_{75,r}$ be the type-7 sample quartiles of $\bar{y}_t$ in record r. We label a day high if $\bar{y}_t \geq q_{75,r}$ and middle if $q_{25,r} \leq \bar{y}_t < q_{75,r}$. Lower-quartile days do not enter the contrast. There were no threshold ties; every month-year contains eight high days and 14 or 15 middle days. These are relative upper-quartile days: the threshold is redefined within each record, and no extreme-value tail is involved.

The two groups are compared at each scale by the relative difference of their dispersion means. Let $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ be the high- and middle-day sets in record (y, m), and let $\overline{Q}^E_{ymh}$ and $\overline{Q}^M_{ymh}$ denote the corresponding graph-dispersion means. The scale-specific relative contrast and its summer average are

$$R_{ymh} = \frac{\overline{Q}^E_{ymh}}{\overline{Q}^M_{ymh}} - 1, \qquad \widehat{\theta}_y = \frac{1}{3H} \sum_{m=1}^{3} \sum_{h=1}^{H} R_{ymh}. \tag{4}$$

For a set $\mathcal{Y}$ of summers, the primary finite-record estimand is

$$\widehat{\theta}_{\mathcal{Y}} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \widehat{\theta}_y. \tag{5}$$

A negative value means that high-regional-mean fields vary less across space. Because the bandwidths are equally spaced on the log scale, the arithmetic mean does not privilege any single bandwidth within the prespecified grid. It is a discrete five-scale summary, not an integral over all spatial scales. The log-ratio sensitivity replaces R_{ymh} by $\log(\overline{Q}^E_{ymh}/\overline{Q}^M_{ymh})$ at the same three stages of averaging and back-transforms the final mean for interpretation. A second exploratory sensitivity uses the bounded symmetric contrast $2(\overline{Q}^E_{ymh} - \overline{Q}^M_{ymh})/(\overline{Q}^E_{ymh} + \overline{Q}^M_{ymh})$. The three effect measures are applied to identical fields and labels in the denominator stress test.

2.2 Exact spatial and structural decomposition

Graph dispersion accumulates over edges. We allocate it to sites by splitting every undirected edge equally between its endpoints; the allocation is exact and symmetric, though not unique. For location i, define

$$q_{ih}(\mathbf{y}) = \frac{1}{4S_h} \sum_{j \neq i} w_{ij,h} (y_i - y_j)^2. \tag{6}$$

The identity is exact because the double sum counts every edge twice:

$$\sum_{i=1}^{n} q_{ih}(\mathbf{y}) = \frac{1}{2S_h} \sum_{i<j} w_{ij,h} (y_i - y_j)^2 = Q_h(\mathbf{y}). \tag{7}$$

The allocation is translation invariant and nonnegative for a single field. It also respects the graph geometry: a site receives contributions only from its incident weighted contrasts, while distant pairs carry little weight in a local graph. Equal edge splitting avoids assigning an arbitrary direction to an undirected spatial relationship. Let $\overline{q}^E_{iymh}$ and $\overline{q}^M_{iymh}$ denote the high- and middle-day means within record (y, m), and let $\overline{Q}^M_{ymh} = \sum_i \overline{q}^M_{iymh}$. The site contribution to the record-level relative effect is

$$c_{iymh} = \frac{\overline{q}^E_{iymh} - \overline{q}^M_{iymh}}{\overline{Q}^M_{ymh}}, \qquad \sum_i c_{iymh} = R_{ymh}. \tag{8}$$

Unlike q_{ih}, c_{iymh} may be negative because it measures a change between event regimes. Its denominator is the record-specific middle-day dispersion. Sites in months with different baseline variability therefore enter the final allocation on the same relative scale as the corresponding record-level estimator.

Let $\mathcal{Y}$ be the set of summers. With $H = 5$, the reported node allocation is

$$c_i = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \frac{1}{3} \sum_{m=1}^{3} \frac{1}{H} \sum_{h=1}^{H} c_{iymh}, \qquad \sum_{i=1}^{n} c_i = \widehat{\theta}_{\mathcal{Y}}. \tag{9}$$

The order and weights in Equation (9) match the primary estimator: equal months within a summer, equal graph scales, and equal summers. The identity is checked numerically after every analysis. Changing the spatial support changes the incident edges and hence the allocation. We therefore calculate every main-text map from the primary 121-node graph. For a scale-specific map, we retain one h in Equation (9) instead of averaging over H; the observed-node sum then recovers the corresponding scale-specific contrast.

For display, we fit a low-rank thin-plate REML surface to each set of 121 observed-node values and evaluate it on a fine land-masked raster. Graph construction, estimation, uncertainty calculations and node-sum identities all use the observed nodes, which are overlaid on each map. A negative c_i allocates part of the regional contraction to edges incident to site i; the allocation is descriptive at the coordinate level. We also map the regime means of $y_{it} - \bar{y}_t$ and their difference to separate spatial-pattern change from a shift in the daily regional mean.

To quantify the dominant north–south structure, let $\boldsymbol{\ell}$ be centred latitude and $\mathbf{z}_t = \mathbf{y}_t - \bar{y}_t \mathbf{1}$. Define

$$\widehat{\beta}_{th} = \frac{\boldsymbol{\ell}^\top \mathbf{L}_h \mathbf{z}_t}{\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}, \qquad \mathbf{e}_{th} = \mathbf{z}_t - \widehat{\beta}_{th} \boldsymbol{\ell}.$$

Orthogonality in the $\mathbf{L}_h$ inner product gives

$$Q_h(\mathbf{y}_t) = \frac{\widehat{\beta}_{th}^2 \boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}{2S_h} + \frac{\mathbf{e}_{th}^\top \mathbf{L}_h \mathbf{e}_{th}}{2S_h}. \tag{10}$$

High-minus-middle means of the two terms, divided by middle-day total dispersion, add exactly to R_{ymh}. This exact energy accounting is descriptive. More generally, for a centred spatial basis matrix $\mathbf{B}$, the Laplacian projection has coefficients $\widehat{\boldsymbol{\beta}}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^- \mathbf{B}^\top \mathbf{L}_h \mathbf{z}_t$, where $(\cdot)^-$ is a generalised inverse. Centred latitude is the primary structural summary. Nested sensitivities add longitude and then elevation, obtained by dividing the official invariant ERA5-Land geopotential field by standard gravity. We compare the total structured and residual energies of these nested column spaces; no order-specific latitude, longitude or elevation contribution is assigned.

We also separate the persistent monthly field from daily departures. Write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$, where $\boldsymbol{\mu}_m$ is the site-specific 35-summer mean field for month m. Then

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \tag{11}$$

The first term is constant within a month and cancels from the high-minus-middle numerator. The other two terms quantify change in anomaly energy and change in climatology–anomaly alignment. Dividing both differences by the middle-day total dispersion yields two components that add exactly to R_{ymh}.

3 Finite-record inference and analysis protocol

3.1 Product cyclic-shift randomisation

Days within a month are serially dependent, and a month holds only 30 or 31 of them, so randomisation proceeds month by month rather than day by day. For a given month-year record, we rotate the complete daily outcome profile by a randomly chosen offset, while the mean-WBT series and its labels remain untouched. Rotating the whole profile rather than individual days keeps the dependence between bandwidths intact, and the statistic is recomputed after every shift. The one-sided Monte Carlo p-value is

$$p = \frac{1 + \sum_{b=1}^{B} I(T_b^{\text{shift}} \leq T_{\text{obs}})}{B+1}. \tag{12}$$

The validity of this scheme rests on cyclic invariance, an explicit condition rather than an implication of generic stationarity. For record r, let $X_r = (\bar{y}_{rt})_{t=1}^{n_r}$ denote the mean series and let D_r collect the H graph-dispersion values for each day, with $X = (X_1, \ldots, X_R)$ and $D = (D_1, \ldots, D_R)$. The operator C_{n_r} rotates all H columns of D_r together, and $G = \prod_r C_{n_r}$ is the group of all joint rotations. The null hypothesis is group invariance:

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \tag{13}$$

Proposition 1 (Product cyclic-shift validity)**.** *Let G be finite and suppose Equation (13) holds. For a measurable lower-tail statistic $T(X, D)$ whose definition is fixed before the transformations are examined, define*

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \le T(X, D)\}.$$

Then, for every $u \in [0, 1]$,

$$\Pr\{p_G(X, D) \le u \mid X\} \le u \quad \textit{almost surely.}$$

If the $|G|$ transformed statistics have no ties, p_G is uniform on $\{1/|G|, \ldots, 1\}$ conditional on the invariant orbit, and the enumeration test is exact at its attainable levels. If B transformations are sampled independently and uniformly from G, independently of (X, D), the plus-one value p_B in Equation (12) satisfies

$$\Pr\{p_B \le u \mid X\} \le \frac{\lfloor (B+1)u \rfloor}{B+1} \le u.$$

Proposition 1 is a finite-sample result: no independence between days is needed. The conditional statement holds even with ties or repeated group actions, although exact uniformity on the attainable grid requires the no-tie condition stated above. The full proof is in Supplementary Section S1, and the Monte Carlo version follows the argument of Phipson and Smyth (2010).

An alternative family of time-shift tests avoids the wrap-around assumption by truncating the series and needs only strict stationarity (Harris, 2020; Yuan and Shou, 2024). In its conservative finite-sample form, such a test requires at least 19 shifts on each side to reach a 0.05 cutoff. Our monthly records have 30 or 31 days, which is too short to satisfy that requirement while leaving a usable aligned window. We therefore use the cyclic test and state its stronger invariance assumption explicitly.

For scale-specific inference in the development records, each shifted profile contributes its minimum relative effect across the five bandwidths, and an observed scale contrast is compared with the distribution of these minima. Under the complete joint null, this single-step comparison controls the family-wise error rate in the weak sense; strong control need not hold under arbitrary partial nulls. Because every component is a unitless relative change, taking the minimum does not mix quantities measured on different scales.

A second uncertainty summary is built from the six month-year contrasts: their unweighted mean, with a 95% interval formed from a t_5 critical value and the between-record standard error. We use this six-record interval only as a descriptive between-record summary.

3.2 Development and evaluation design

We fixed the multiyear specification before acquiring or inspecting the remaining ERA5-Land summers. The design reserves 2015 and 2022 for method development and applies Equations (4)–(5) to the 33 evaluation summers without re-estimating labels, bandwidths or field-selection rules. The primary target $\widehat{\theta}_{\mathcal{Y}_H}$ describes those 33 summers and requires no stationarity across years.

For the global cyclic-shift analysis, set $T_{\text{obs}} = \widehat{\theta}_{\mathcal{Y}_H}$. Each Monte Carlo transformation rotates the five-column dispersion profile within each of the 99 evaluation month-year records, then recomputes every record ratio and the equal-month, equal-scale and equal-summer mean. We use $B = 99{,}999$.

Effect size is reported through the record mean, summer-specific contrasts, negative-summer count and leave-one-summer-out range. Supplementary Table S1 separates the primary estimator from exploratory analyses. We specified the cyclic-shift analysis and these exploratory analyses after observing the 33-summer estimate. They provide conditional assessments of the observed record rather than prospective confirmation or inference on a long-run climate parameter. The exploratory analyses include alternative effect measures, the historical extension, spatial refinements, a dense bandwidth curve, nested spatial bases and the climatology–anomaly decomposition.

3.3 Process-level reference

A secondary interpretation treats the summer sequence as a stationary weakly dependent process. Under standard strong-mixing, moment and long-run variance conditions, its mean admits a central-limit approximation; a growing-bandwidth HAC estimator needs a stronger uniform autocovariance condition (Ibragimov and Linnik, 1971; Newey and West, 1987; Andrews, 1991). The full statement, ratio negative-moment conditions and proof are in the Supplement. At 33 summers, simulation shows that Student and HAC intervals can have substantial undercoverage under serial dependence. We report them as independent-summer or process-model references, separate from the finite-record randomisation analysis.

4 Simulation study

The simulations examine day-level cyclic randomisation and the behaviour of process-model intervals at the 33-summer record length. These are separate sampling problems: additional days within a month do not increase the number of summers. Table 1 specifies both designs.

4.1 Data-generating processes

The day-level experiment uses the observed coordinates and six 30–31-day records. Its circular Gaussian null satisfies Proposition 1; two finite-autoregression variants deliberately violate wrap-around invariance. Each Gaussian anomaly is projected through $\mathbf{I} - \mathbf{1}\mathbf{1}^\top/n$, so its sample mean across the 121 sites is exactly zero and the field mean equals the series used to form labels. The centred anomaly is independent of that mean. This calibration design isolates the randomisation properties; the fixed-hour and anomaly-field analyses in Section 5 examine field selection and label–field dependence in the observed record. Amplitude reduction, correlation-range expansion and gradient suppression produce true profile contractions of 5.9%–27.4%, calculated from the known quadratic forms. The Monte Carlo standard error is 0.7 percentage points at 5% rejection and 1.6 points at 50% power, so differences of one or two points do not support method rankings.

Seven prespecified procedures are compared in the day-level experiment: the proposed mean of five graph-dispersion contrasts, ordinary spatial variance, a four-nearest-neighbour semivariance, a classical binned semivariance using pairs 400–600 km apart (Cressie, 1993), an equal-weight profile of five semivariance contrasts formed from prespecified pair-distance quintiles, and global Moran's I and Geary's C (Moran, 1950; Geary, 1954) with the 503-km Gaussian weights. The five dispersion quantities use lower-tail tests; because the direction of an autocorrelation change depends on the mechanism, Moran and Geary use the absolute high-minus-middle difference and a two-sided group test. The single distance bin is a local benchmark; the five-bin profile is the direct variogram-profile competitor. Specifically, the 7,260 pair distances are split at their prespecified quintiles into sets $\mathcal{B}_k$, and

$$\Gamma_k(\mathbf{y}) = \frac{1}{2|\mathcal{B}_k|} \sum_{(i,j)\in\mathcal{B}_k} (y_i - y_j)^2.$$

The profile's statistic applies Equation (4) to each Γ_k, with the same equal-record, equal-bin averaging as the graph statistic. All metrics are evaluated on identical simulated fields with the same joint shifts. The graph profile avoids hard bin boundaries and admits the exact node allocation in Equation (9); we compare calibration and power to assess whether these properties entail a loss relative to the six competitors.

The repeated-summer experiment compares the Student interval, fixed lag-2 calculation, Bartlett estimator with $L_R = \lfloor R^{1/3} \rfloor$, sign test and three-test rule. The -7% alternative is close to the evaluation-record estimate.

4.2 Calibration and comparison across spatial mechanisms

Table 2 reports rejection rates over the nine spatial alternatives; Figure 1 shows how each method's power range depends on the unknown mechanism.

Across eight cyclic-invariant designs, graph-profile rejection ranged from 4.5% to 6.8%, while the six competitors spanned 4.0%–6.4%. The graph test had rejection rates of 4.5% at temporal correlation 0.75 and 6.1%

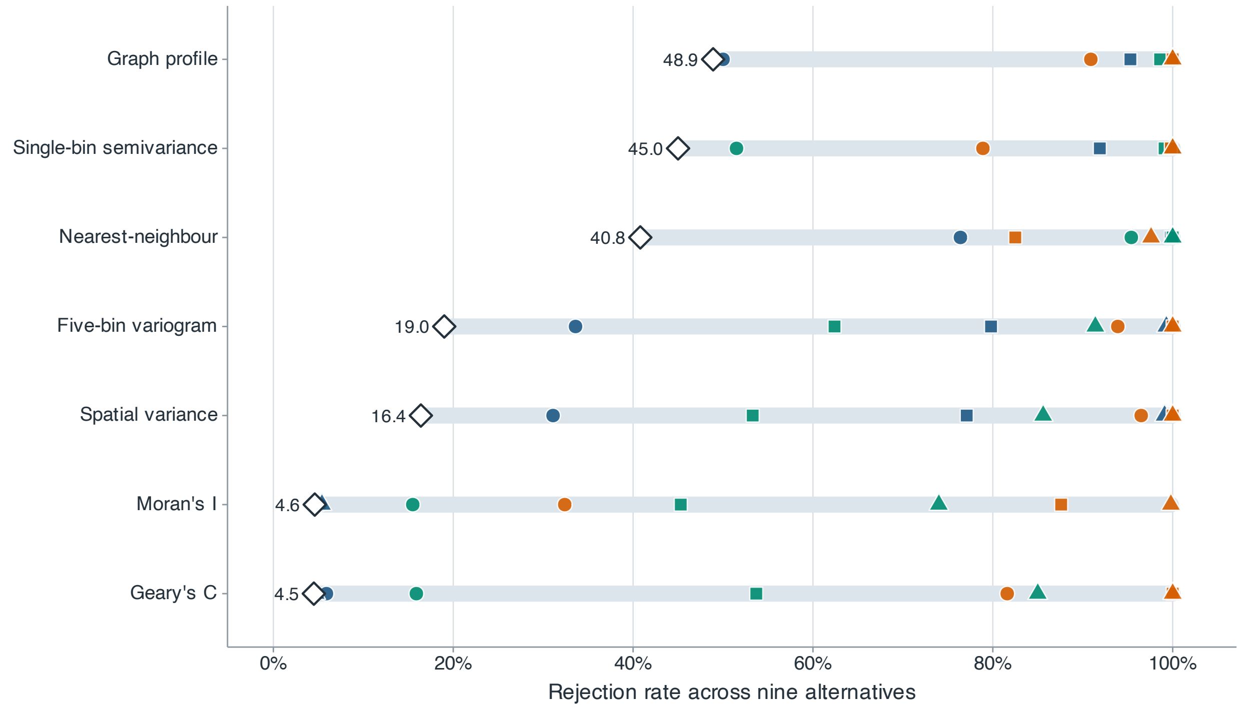

Figure 1: Power across nine simulated spatial alternatives. Blue, green and orange marks denote amplitude reduction, correlation-range expansion and gradient suppression; circles, squares and triangles denote weak, medium and strong changes. For each method, the pale segment spans power across the nine alternatives and the diamond marks the minimum.

Alt text: A range-and-dot plot compares seven methods. Each method has nine coloured and shaped points, a pale segment spanning their range and a diamond at the minimum; the ranges overlap substantially.

under the t_3 field. The two non-circular boundary cases gave 4.6% and 5.1%; these results do not extend Proposition 1 beyond its invariance condition.

Gaussian-alternative absolute bias was below 0.48 percentage points and root mean squared error was 2.6–3.8 points. Under the original t_3 null, graph randomisation size was 6.1%. We then ran a paired denominator stress test with 2,000 heavy-tailed data sets under each of the null and -7% alternatives and 999 common product shifts per data set. Null rejection was 4.70%, 4.90% and 4.75% for the raw ratio, log ratio and bounded symmetric contrast. On their native scales, the corresponding null absolute biases were 0.372, 0.070 and 0.066, and root mean squared errors were 1.472, 0.328 and 0.277. Under the -7% alternative, root mean squared errors were 1.369, 0.328 and 0.275, while rejection was 7.95%, 8.50% and 7.90%. The bounded contrast sharply reduced estimator dispersion but did not materially increase power in this design. Randomisation calibration can therefore coexist with unstable effect estimation; in this design, resistant summaries reduce but do not eliminate the problem.

No method dominates Table 2. Nearest-neighbour semivariance is strongest for range changes, but its power is only 40.8% under weak gradient suppression, compared with 90.9% for the graph profile and 96.5% for variance. The five-bin variogram is similar to the graph profile for gradient suppression (93.9% versus 90.9% at the weakest setting), but is less sensitive to the weak amplitude and range changes (33.6% and 19.0% versus 50.0% and 48.9%). Moran's I and Geary's C have near-nominal power for pure amplitude changes because standardisation removes amplitude. The profile's minimum power across the nine tabulated alternatives is 48.9%, higher than the minimum of each competitor in this experiment. Its advantage is coverage across unknown mechanisms, not pointwise dominance.

The scale-specific profiles retain information lost by a scalar test. Strong amplitude reduction changes all three summaries by about 19%; range expansion acts most strongly on the local graph (28.2%), while gradient suppression acts on variance and the broad graph (43.9% and 42.5%) but changes the local graph by only 9.3%.

The baseline independence design was supplemented by eight simulation scenarios in which the label-generating mean and spatial field share structure (full specifications are in the Supplement). In the shared-latent null, the sign of the spatial gradient has mean correlation 0.61 with regional WBT, but its squared energy is constant, so the true dispersion contrast is zero. The other nulls add a within-month trend, anisotropic spatial covariance or selection of a daily peak from 24 simulated hourly regional means, with hour-invariant anomaly covariance. Corresponding alternatives attenuate the gradient, impose a seasonally declining gradient or reduce high-day anomaly amplitude. Graph null rejection is 4.0%–7.0%, compared with 4.3%–7.3% for the five-bin variogram. These results are close to the cyclic-invariance calibration but fall outside Proposition 1. The graph profile is at least as powerful in three of four alternatives, whereas the variogram is slightly stronger for the seasonal-gradient alternative. Smooth graph weighting and exact allocation therefore show no consistent power penalty across these scenarios.

4.3 Process-reference stress test

At $R = 33$, increasing Gaussian year correlation from 0 to 0.6 reduces Student coverage from 95.2% to 66.1%, fixed-lag coverage from 93.3% to 78.7%, and growing-HAC coverage from 91.2% to 79.0%. Corresponding Gaussian null rejection rises to 21.2%, 14.4% and 14.1%. The error is driven by long-run variance estimation rather than centring. None of the three process-model intervals is reliable as primary evidence at this record length under the simulated dependence. Full results for 108 cells, including skewed and heavy-tailed innovations, appear in the Supplement.

An additional 10,000-replication experiment uses the exact 1991–2025 calendar and removes 2015 and 2022. The largest absolute mean bias over correlations 0, 0.3 and 0.6 and effects 0 and -7% is 0.03 percentage points. Fixed deletion therefore has negligible first-order centring bias in this experiment, but short-record dependence remains: at correlation 0.6, null coverage is 68.3% for the independent-summer interval and 80.2% for calendar lag 2.

5 Humid-heat application

We apply graph dispersion to synchronous wet-bulb temperature fields in eastern China. The estimand, labels, scales and decision rule were fixed using 2015 and 2022; the record estimate uses the other 33 evaluation summers. The ERA5-Land analysis panel covers every summer in the specified 1991–2025 primary window. We apply the same construction separately to 1950–1990.

5.1 Data construction and analysis design

ERA5-Land supplies hourly land-surface fields on a 0.1° latitude–longitude grid (Muñoz-Sabater et al., 2021). We use 2-m air temperature, 2-m dew-point temperature and surface pressure for June–August 1991–2025 over 105–125°E and 20–42°N. Both limits were fixed during method development. The rectangle provides a regular support spanning the subtropical-to-temperate north–south gradient of eastern China while excluding most of the Tibetan Plateau west of 105°E. Its northern and southern limits bound that intended gradient, and the 1991 start reflects the original data acquisition rather than a climatic breakpoint. We analysed the 33 evaluation summers only after fixing the specification.

Before accessing the earlier values, we recorded a separate extension protocol. It retains the 121 sites, Bolton WBT and quality-control rules, UTC regional-peak fields, within-record type-7 labels, five physical bandwidths and the equal-month, equal-scale and equal-summer estimator. The extension covers all 41 June–August seasons from 1950, the first year of the ERA5-Land record, through 1990. The protocol also specifies separate summaries for 1950–1978 and 1979–1990 because the earlier segment uses the preliminary ERA5 back extension (ECMWF production documentation). This split follows the production history rather than the observed effect.

Every 16th longitude and 18th latitude cell from a fixed north-west origin gives a 13×13 candidate lattice; removing ocean cells leaves 121 sites (Figure 2). The 35 summers contain 9,350,880 quality-controlled hour–site records and 92 complete UTC fields per summer. Because of NetCDF packing, 2,092 dew-point values

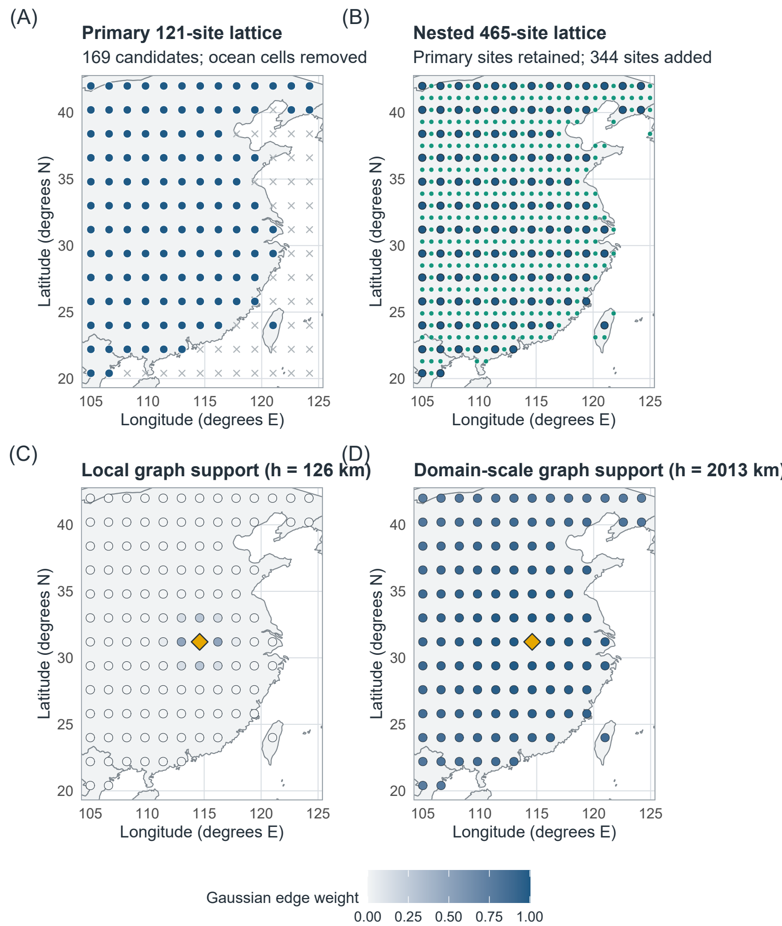

Figure 2: Spatial sampling and graph support. (A) Primary lattice with 121 land sites from 169 candidates. (B) Nested lattice adding 344 land sites while retaining every primary site. (C,D) Gaussian edge weights from one reference site under the 126- and 2,013-km graphs, showing how bandwidth changes the geographic support at fixed coordinates. Basemap: Natural Earth 1:50m, accessed through the R `maps` package.
Alt text: Four eastern-China maps compare the primary and nested site lattices, then show weights concentrated near one reference site under the 126-km graph and spread across the domain under the 2,013-km graph.

exceeded air temperature slightly (0.022% of records); these pairs were clipped at equality before WBT was calculated. The primary target is a fixed-site average. For the area-weighted sensitivity, $a_i = \cos(\text{latitude}_i)$,

$$\bar{y}_t^{(a)} = \frac{\sum_i a_i y_{it}}{\sum_i a_i}, \qquad Q_{h,a}(\mathbf{y}) = \frac{\sum_{i<j} a_i a_j w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} a_i a_j w_{ij,h}}.$$

We first replace Q_h by $Q_{h,a}$ with the original labels fixed, then redefine the labels from $\bar{y}_t^{(a)}$ at the primary peak hours. The two calculations separate spatial weighting from event relabelling.
WBT is calculated from temperature T and dew point T_d in kelvin and surface pressure p in pascals. Vapour pressure is $e(T_d) = 611.2\exp\{17.67(T_d - 273.15)/(T_d - 29.65)\}$ Pa, and the Bolton mixing-ratio, lifting-condensation-temperature and equivalent-potential- temperature relations are evaluated in these units (Bolton, 1980). Saturated T_w is obtained by solving

$$\theta_e(T_w, T_w, p) = \theta_e(T, T_d, p) \tag{14}$$

at the original pressure with 40 bisection iterations on $[T_d, T]$; even a 100-K initial interval then falls below 10^{-9} K. Values with $T_d > T$ are clipped at equality before evaluation. The Stull approximation (Stull, 2011) is a sensitivity analysis; the primary calculation follows the pseudoadiabatic definition discussed by Davies-Jones (2008).

For each UTC day, the 121 simultaneous values at the peak regional mean form the daily field. Within each month-year, the upper regional-mean quartile is high, the middle half is middle and the lower quartile is excluded. Neither the label nor the selected hour uses graph dispersion. UTC+8 days, alternative thresholds, sitewise maxima, all 24 fixed UTC hours and daily-mean fields provide sensitivity checks. At each fixed hour, both the original peak-based labels and labels recomputed from that hour's regional mean are analysed; daily-mean labels are likewise recomputed from daily-mean WBT. A dated event manifest records the type-7 thresholds, membership and absence of ties.

Three checks address within-month seasonal progression. We compare the day-of-month distributions of the two regimes, fit and remove a separate linear day-of-month trend at every site within each year–month, and subtract each site's calendar-day climatology, estimated from all summers except the one being analysed. Event labels remain unchanged in the last two checks, so they ask whether the original regime contrast survives removal of those calendar components. A separate date-matched calculation compares each high day with all middle days in the same month-year lying within ±3 or ±5 calendar days. It first averages the available high-day-specific ratios and then uses the same equal-month, equal-scale and equal-summer hierarchy as the primary estimator; unmatched-day and empty-record counts are retained.

An exploratory continuous diagnostic regresses $\log Q_h$ on centred regional mean WBT within each month-year and bandwidth. Slopes are then averaged equally over months, bandwidths and evaluation summers. This slope supplements the quartile estimand with a graded summary.

Alternative graph kernels are $\exp(-d_{ij}/\rho)$ and $[1 - d_{ij}/\rho]_+^2$. For each Gaussian scale, ρ is found by bisection so that the alternative kernel has the same edge-weighted mean pair distance. Across the five scales, their effective edge counts range from 334 to 7,112 and from 382 to 7,113, compared with 366 to 7,109 for the Gaussian graphs; the full matched-distance and effective-edge metadata accompany the analysis.

A nested $0.8° \times 0.9°$ lattice adds 344 land sites while retaining all primary locations. For numerical convergence, we hold the five physical bandwidths, primary peak times and labels fixed, and compare 121 sites with a 239-site longitude refinement, a 236-site latitude refinement and all 465 sites. We also reselect peak times and labels on the full grid as a separate design sensitivity. Thin-plate REML surfaces (Wood, 2003) are used only for display in Figures 3 and 4; every estimate uses observed nodes.

We assess ERA5-Land WBT against the National Oceanic and Atmospheric Administration Integrated Surface Database (NOAA ISD) (Smith et al., 2011). We selected ten summers excluded from method development (1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025) before evaluating their station values. After applying the archive-availability and station-operating-period criteria described in the Supplement, a deterministic maximin design selected 30 dispersed stations. We retained a station-year when it had at least 400 quality-controlled, complete exact-hour observations of temperature, dew point and sea-level pressure; station elevation supplied surface pressure under a standard atmosphere. At two offshore stations, all variables were missing from the nearest ERA5-Land grid cells, leaving 28 usable stations and 175,172 exact-hour WBT pairs.

For the station-field calculation, we kept the 121-site ERA5-Land peak times and high–middle labels fixed. At each retained time, NOAA and ERA5-Land graph contrasts use the identical station subset and require at least 10 stations. All 30 year-month records retained at least one field in each regime. The 503-, 1,006- and 2,013-km graphs were designated as station-supported before evaluation; the two narrower graphs diagnose the limit of the station geometry. This analysis assesses measurement agreement and whether the sparse station network yields the same direction of spatial contrast under the fixed ERA5-Land labels.

5.2 Primary evaluation-period multiscale result

Across the 33 evaluation summers, the prespecified five-scale finite-record contrast was −7.28%. None of the 99,999 independently sampled product cyclic shifts produced a statistic as negative as the observed value, giving the plus-one one-sided value $p = 0.00001$ under the conditional invariance null, the minimum attainable value for this Monte Carlo run. The complete five-scale daily profile was shifted jointly within each of the 99 month-year records. This result assesses the observed finite record under the stated invariance condition.

Twenty-eight summer-specific contrasts were negative, and leave-one-summer-out estimates ranged from −7.69% to −6.64%. For process-model comparison, the 95% independent-summer t-reference interval is $[-10.00, -4.57]$%. The fraction negative is 0.85 (Wilson interval $[0.69, 0.93]$), with a one-sided independent-

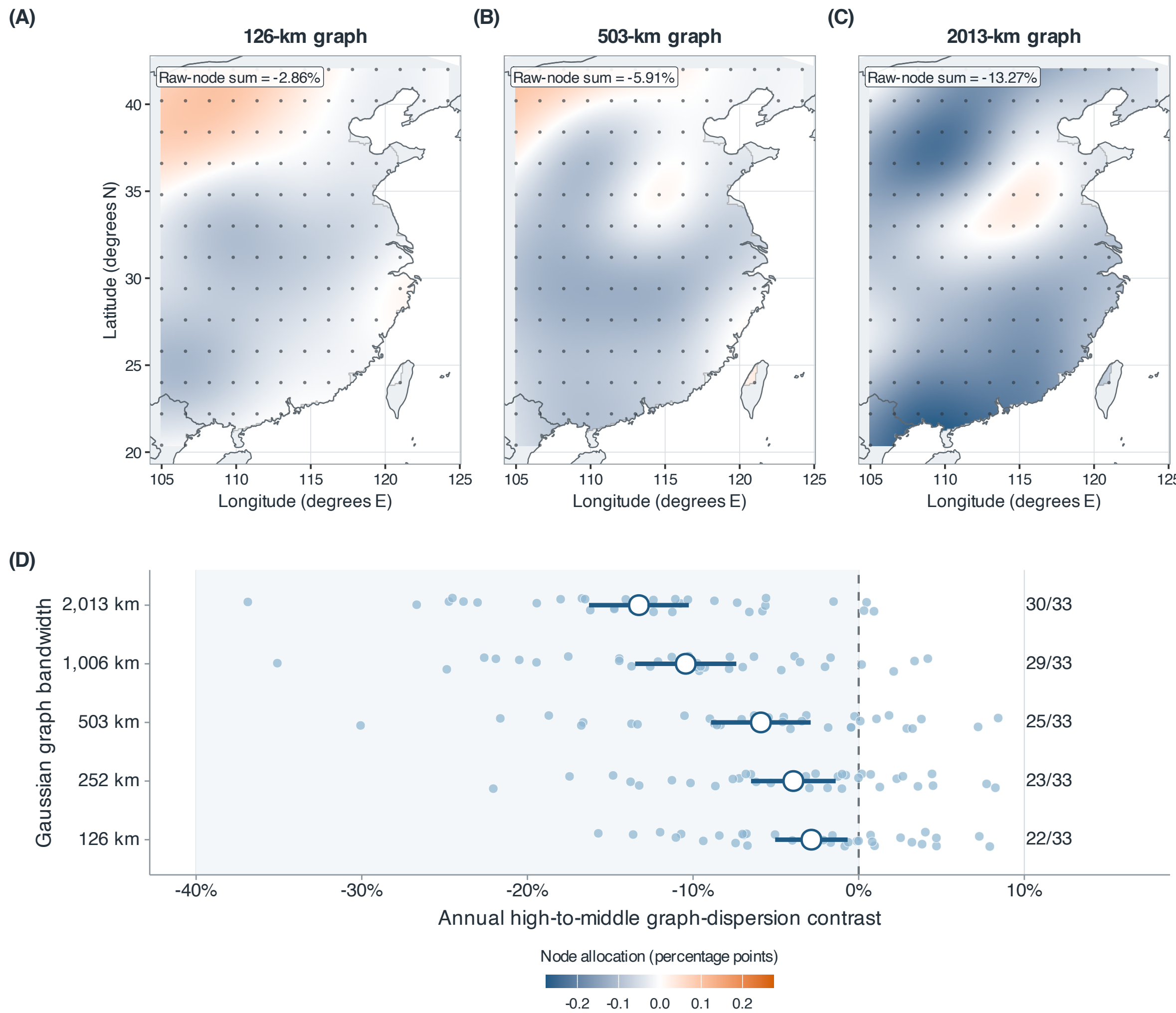

Figure 3: Scale-specific node allocations and summer contrasts for the 33 evaluation summers. (A–C) Allocations for the 126-, 503- and 2,013-km graphs share a symmetric colour scale; their observed-node sums are −2.86%, −5.91% and −13.27%. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes. (D) Small circles show summer-specific contrasts at all five bandwidths; large circles and bars show means and 95% descriptive intervals. Labels give the numbers of negative summers.

Alt text: Three smoothed eastern-China maps, each overlaid with the 121 analysis sites, show increasingly coherent negative node allocations at 126, 503 and 2,013 km. A lower panel shows the 33 annual contrasts at all five bandwidths, with mean contrasts and negative-summer counts increasing in magnitude with bandwidth.

sign reference value of 3.3×10^{-5}. These process-model summaries are secondary; the cyclic-shift result provides the finite-record inferential assessment.

The fitted trend was −0.22 percentage points per year (95% descriptive OLS reference interval $[-0.48, 0.05]$, two-sided $p = 0.10$). The 2008–2025 mean was 2.95 points more negative than the 1991–2007 mean, with a descriptive Welch reference interval of $[-8.43, 2.52]$ points. These descriptive trends are too imprecise to establish temporal stability; process-level inference requires assumptions beyond the finite record.

Effects increase in magnitude from −2.86% under the 126-km graph to −13.27% under the 2,013-km graph (Table 3; Figure 3). Negative-summer counts rise from 22 to 30. A 31-point log-spaced bandwidth curve decreases monotonically across the same 126–2,013-km range. Complete-graph variance falls 14.45% and is negative in all 33 summers.

Historical extension, 1950–1990. Applying the unchanged construction to 1950–1990 gave a five-scale contrast of -11.04% (95% descriptive interval $[-12.84, -9.24]\%$). Forty of 41 summer contrasts were negative, the leave-one-summer-out range was -11.33% to -10.62%, and none of 99,999 product shifts was as negative as the observed statistic (plus-one $p = 0.00001$). The scale profile was $-5.96, -7.51, -10.08, -14.50$ and -17.17% from 126 to 2,013 km, with 38, 39, 40, 41 and 41 negative summers. The negative sign and strengthening contraction with bandwidth also appear in the 41 summers preceding the primary window.

The production-history split gave -10.50% in 1950–1978 (28 of 29 summers negative) and -12.34% in 1979–1990 (12 of 12 negative). We report these segments because the pre-1979 record uses different forcing; the split is descriptive. At 2,013 km, the historical anomaly-energy component was -0.28 points (95% descriptive interval $[-1.01, 0.45]$), whereas the climatology–anomaly cross component was -16.89 points ($[-19.31, -14.47]$) and negative in all 41 summers. Both the latitude and latitude–longitude structured components were negative in every summer at every bandwidth. The earlier record is therefore consistent with the same broad structural pattern, subject to the documented pre-1979 forcing caveat.

5.3 Spatial structure and energy decomposition

The decomposition in Equation (10) attributes most of the contrast to the north–south gradient. At 126 km it accounts for -2.58 points (95% independent-summer reference interval $[-2.92, -2.24]$) of the -2.86-point contrast. From 252 to 2,013 km its mean contribution is more negative than the total ($-4.60, -8.38, -12.73, -14.96$ points), while residual contrasts contribute $+0.66, +2.48, +2.29, +1.69$ points. The gradient component is negative in all 33 evaluation summers at every bandwidth. Adding longitude to the basis makes the 2,013-km structured component -16.23 points, compared with -14.96 points for latitude alone; the corresponding residuals are positive. The latitude–longitude–elevation basis gives a 2,013-km structured component of -14.34 points (95% descriptive interval $[-16.75, -11.92]$), negative in all 33 summers, and a residual of $+1.07$ points. At 126 km, however, its structured component is -1.16 points ($[-2.66, 0.33]$) and is negative in only 18 summers. The nested bases show a broad low-dimensional geographic–topographic structure dominated by the north–south component, with weaker structured attribution at 126 km. Node allocations are negative at 106 sites and sum exactly to -7.28%; they are symmetric incident-edge allocations, not unique local effects.

The climatology–anomaly decomposition in Equation (11) locates the broad-scale change. At 126 km, anomaly-energy change contributes -2.63 points and the climatology–anomaly cross term contributes -0.22 points. At 2,013 km, the anomaly-energy component is $+0.41$ points (reference interval $[-0.50, 1.31]$), whereas the cross term is -13.68 points ($[-16.68, -10.67]$) and is negative in all 33 summers. High-day anomalies therefore tend to oppose the persistent broad geographic contrast, while their own broad-scale energy changes little.

The convergence analysis compares each grid with the 465-site fixed-event reference. The primary 121-site scale profile differs by at most 0.99 percentage points, with a five-scale root mean squared difference of 0.76 points. Latitude refinement reduces these values to 0.50 and 0.38 points; longitude refinement gives 0.93 and 0.73 points. Adding all sites with primary events fixed yields -8.03%; reselecting events yields -7.25%. Contraction strengthens with bandwidth at every resolution. Resolution sensitivity and station limitations are greatest at 126 km, where the primary spacing and station support are coarsest.

The graph effects are relative changes in weighted squared differences. The five-scale mean corresponds to a -3.71% RMS change when transformed as $\sqrt{1 - 0.0728} - 1$; averaging record-specific RMS ratios gives -3.97%.

Threshold, Stull and sitewise-maximum sensitivities range from -6.90% to -8.27%. Early- and late-period summaries have overlapping descriptive intervals. These checks assess the stability of the spatial contrast across analysis choices.

Applying the same equal-month, equal-scale and equal-year formula to 2015 and 2022 gives -15.14%. This descriptive development-period value is excluded from primary inference.

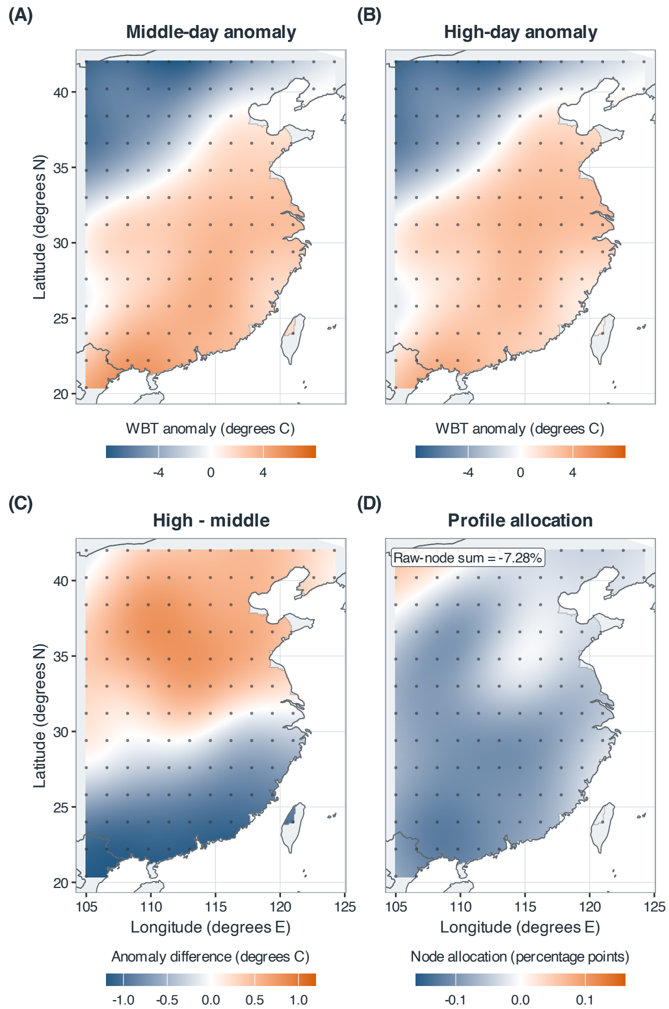

Figure 4: Spatial structure of the primary 121-site evaluation contrast. (A,B) Middle- and high-day within-field anomalies; (C) their difference; (D) exact observed-node allocations averaged over the five prespecified graph scales. The 121 allocations sum to -7.28%, identical to the primary finite-record estimand. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes.

Alt text: Four smoothed eastern-China maps derived from 121 analysis sites, shown as dots. The anomaly surfaces show a weaker north–south contrast on high days; the high-minus-middle surface changes from positive in the north to negative in the south, and most observed-node allocations are negative.

5.4 Robustness and external measurement agreement

The 495 middle-day denominators range from 2.25 to 25.99 $°C^2$ across records and scales; within-scale coefficients of variation are 0.076–0.177, with no near-zero value. The prespecified raw ratio is therefore numerically stable in this application, although the simulation shows that it is not uniformly resistant to small denominators. The back-transformed log-ratio is -8.30%; the bounded symmetric contrast $2(Q^H - Q^M)/(Q^H + Q^M)$ is -8.59%. Cosine-latitude area weighting gives -8.11% when the original labels are fixed, and -7.67% when labels are recomputed from the area-weighted regional mean. Kernels matched to the Gaussian weighted mean distance give -7.71% (exponential) and -7.33% (compact quadratic). All preserve the negative direction and strengthening contraction with bandwidth.

The continuous-intensity diagnostic gives a five-scale $\log Q_h$ slope of -0.065 $°C^{-1}$ (95% descriptive interval $[-0.082, -0.048]$), with negative summer slopes in 31 of 33 years. Scale-specific slopes become more negative from -0.032 $°C^{-1}$ at 126 km to -0.110 $°C^{-1}$ at 2,013 km; the corresponding negative-year counts rise from 29 to 32. The association is therefore graded across the within-month WBT distribution and is not confined

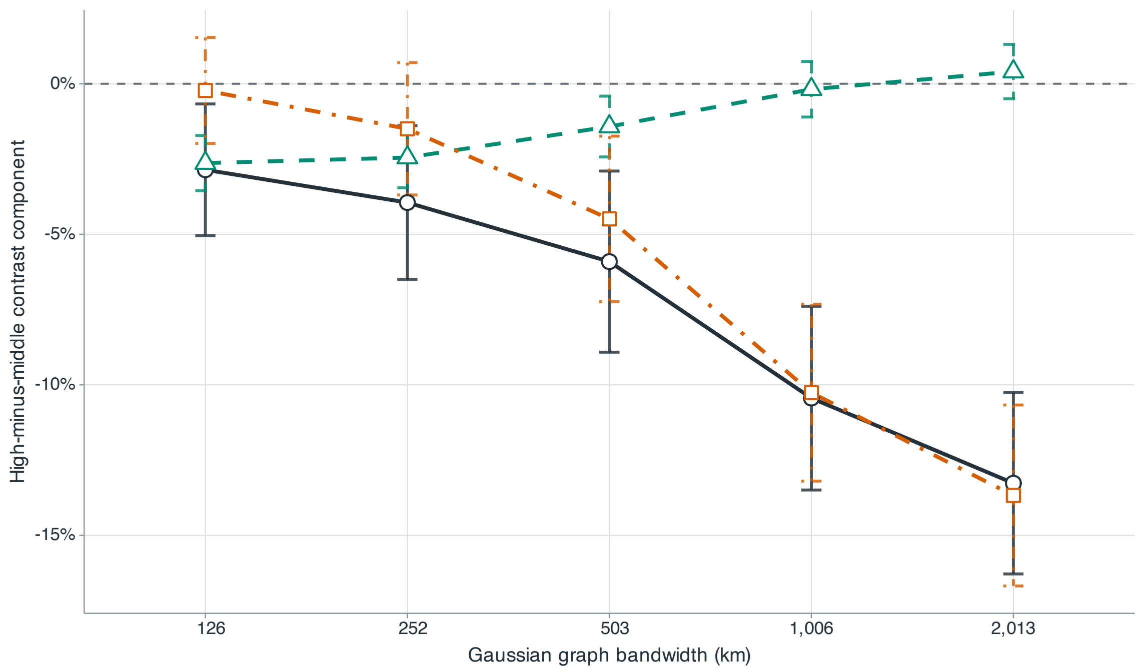

Figure 5: Exact decomposition of the raw-field graph contrast. Dark circles, teal triangles and orange squares denote total change, anomaly-energy change and the climatology–anomaly cross component; bars are 95% descriptive intervals. The fixed monthly-climatology energy cancels within each record.
Alt text: Across five bandwidths, dark circles become increasingly negative. Teal triangles remain near zero at broad scales, while orange squares closely follow the negative total.

to the upper-quartile cut.

The estimated contrast depends on the definition of the spatial field. Subtracting each site's 35-year monthly climatology reduces the estimate to -3.92% ($[-8.67, 0.83]\%$); subtracting the site-year-month mean gives -1.32% ($[-5.63, 2.99]\%$). Site-month standardisation yields -9.87%, but targets a different dimensionless field. Together with the gradient decomposition, these results show that high-day anomalies offset the cross-site climatological gradient, while their own broad-scale energy changes little.

The regime dates have a systematic but nonmonotone seasonal pattern. Averaged over June–August, high days occur 1.96 days later than middle days (95% t-reference interval $[1.10, 2.83]$ days): they occur 7.82 and 6.34 days later in June and July, but 8.29 days earlier in August. Removing a separate linear day-of-month trend within each site–year–month gives -8.81% ($[-13.43, -4.19]\%$); subtracting the leave-one-year calendar-day climatology gives -9.81% ($[-14.22, -5.40]\%$). The negative contrast persists after both seasonal adjustments, although the weaker site-month and site–year–month anomaly results show that different demeaning operations target different spatial fields.

Calendar matching attenuates the effect but preserves its direction. With 95% descriptive intervals in parentheses, the ± 3-day estimator is -5.21% ($[-8.01, -2.41]\%$), with 24 of 33 summer effects negative and 670 of 792 high days matched; the ± 5-day estimator is -5.72% ($[-8.43, -3.02]\%$), with 25 negative summers and 750 matched high days. Every one of the 99 month-year records has at least one eligible match under both windows. The profiles run from -2.32% to -8.87% for ± 3 days and from -2.54% to -9.89% for ± 5 days, retaining the strengthening contraction with bandwidth.

All fixed-hour and daily-mean analyses yielded negative estimates. With original event days fixed, all 24 fixed-hour estimates are negative, from -10.07% at 00 UTC to -6.03% at 11 UTC, and daily-mean fields give -7.58%. When labels are recomputed from each fixed hour, all estimates remain negative, from -14.65% at 22 UTC to -5.42% at 11 UTC; labels recomputed from daily-mean WBT give -9.14%. Peak hours have nearly identical distributions in the original high and middle groups (median 06 UTC). Figure 6 compares the primary estimate with transformation, graph-construction, spatial-resolution and field-definition alternatives.

The station comparison contains 175,172 exact-hour matches at 28 usable stations. ERA5-Land minus station WBT has a bias of -0.263 °C, a mean absolute error of 0.938 °C and a root mean squared error of 1.270 °C.

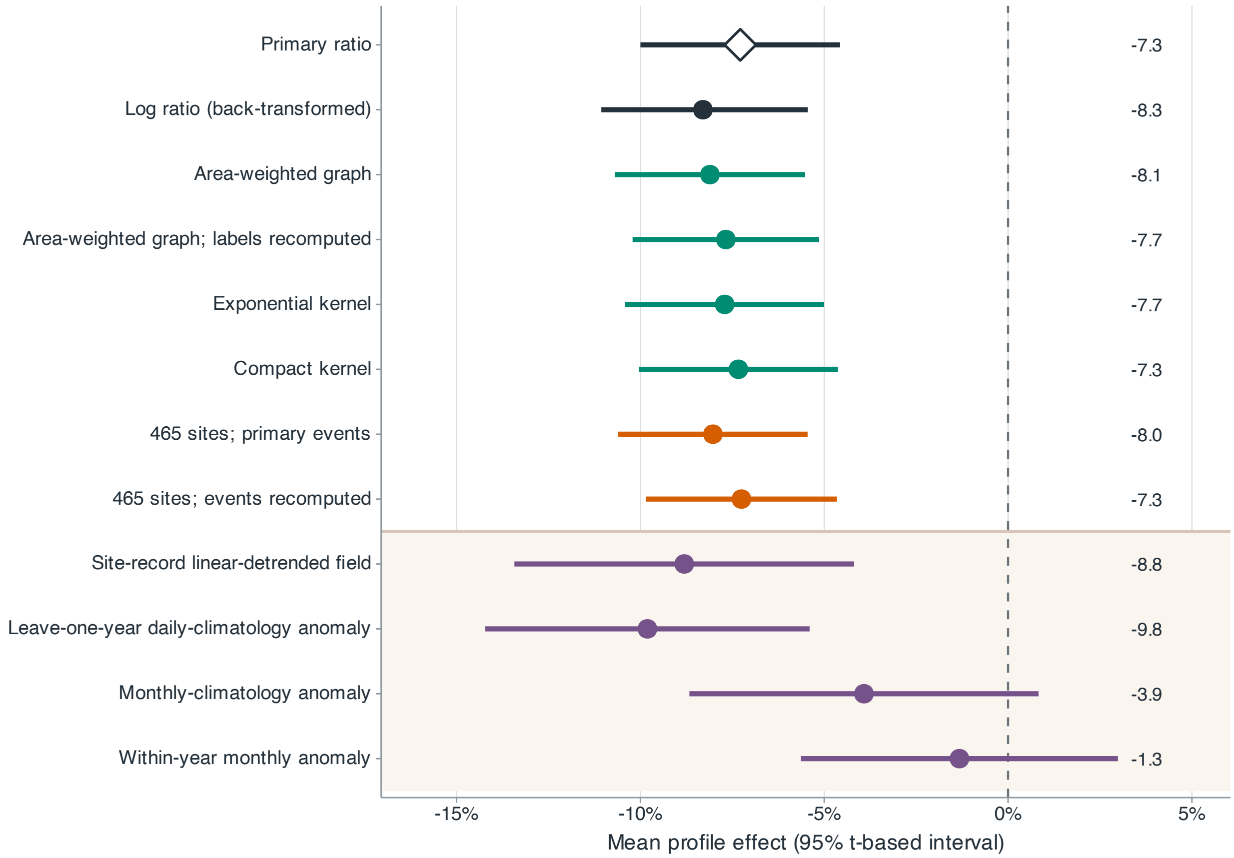

Figure 6: Robustness of the mean profile effect. Dark rows denote the primary and transformation analyses; green, orange and purple rows denote graph construction, spatial resolution and field definition. Points and bars are estimates and 95% descriptive intervals.

Alt text: A forest plot shows mostly overlapping negative estimates across transformation, graph and resolution choices. Two purple-shaded anomaly-field rows lie closest to zero.

The within-station centred correlation is 0.916. The two offshore stations excluded from these summaries had no finite values at their nearest ERA5-Land grid cells.

At 126, 252, 503, 1,006 and 2,013 km, the station high–middle contrasts are -6.23, -7.23, -12.50, -18.03 and -21.47%; the corresponding ERA5-Land values on identical station subsets are -11.60, -12.74, -17.13, -21.60 and -24.23%. The numbers of negative station summers increase across these scales from 6, 7 and 8 to 9 and 9. Averaged over the three predesignated station-supported scales, the station effect is -17.34% ($[-28.49, -6.18]$%), with 9 of 10 summer contrasts negative; the paired ERA5-Land effect is -20.99% ($[-28.57, -13.40]$%), with all 10 negative. The station and matched reanalysis estimates agree in direction at the three designated broad scales. At 126 km, both intervals include zero and provide no clear evidence of a local-scale contrast (Figure 7).

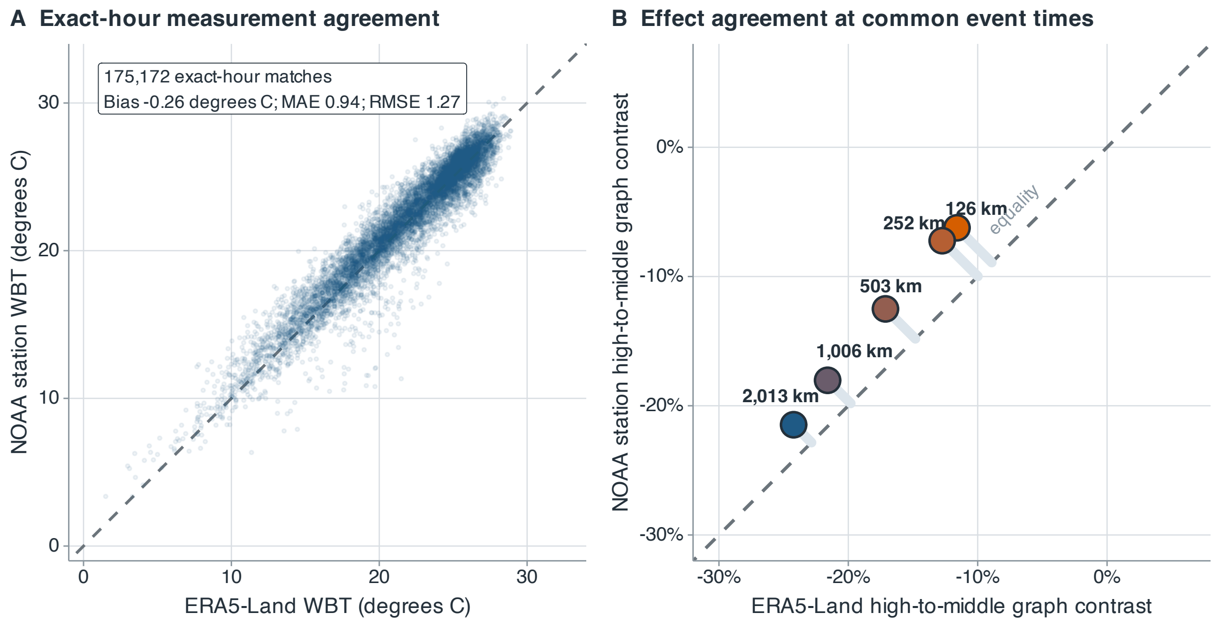

Figure 7: NOAA station and ERA5-Land agreement in ten summers excluded from method development. (A) Exact-hour WBT at 28 stations, with summaries from all 175,172 matches. (B) High–middle graph contrasts at the five fixed bandwidths, using ERA5-Land peak times and labels and identical station subsets. Dashed diagonals denote equality; pale segments in panel B connect station estimates to equality.

Alt text: Two panels compare NOAA stations with ERA5-Land. Exact-hour WBT points in the left panel cluster around the equality line. In the right panel, station and ERA5-Land high–middle contrasts are negative at all five bandwidths and become more negative as the bandwidth increases; station estimates are slightly closer to zero.

6 Discussion

Across the 33 evaluation summers, spatial WBT contrast was lower when the regional mean was high. The five-scale mean was -7.28%, 28 summer effects were negative, and the product cyclic-shift analysis gave $p = 0.00001$ under the conditional invariance null. Contraction strengthened from 126 to 2,013 km, indicating that the largest difference was at broad geographic scales. The 1950–1990 extension showed the same scale pattern, with a mean of -11.04% and 40 of 41 summer effects below zero. Station data also agreed in direction at 503 km and broader: the three-scale mean was -17.34%, with negative contrasts in 9 of 10 summers. At 126 km, the station and matched reanalysis intervals both included zero. These contrasts describe spatial arrangement, a feature distinct from the area exceeding a heat threshold.

The statistical contribution is the integration of scale-indexed graph dispersion, finite-record randomisation and exact spatial accounting for an observational field. The five fixed graphs define a single estimand while retaining its scale profile; product shifts preserve the dependence among scales; and node and energy decompositions recover the regional contrast exactly. This combination shows how the change is distributed over nodes and scales, while keeping the observed-record analysis separate from a stochastic process model. The simulation study supports the randomisation calibration and shows why Student and HAC intervals can be unreliable with only 33 dependent summers.

The decompositions give a specific climatic interpretation. The contrast is dominated by the north–south structure: after removing each day's regional mean, high-day fields are relatively warmer in the north and cooler in the south than middle-day fields. This pattern compresses the persistent monthly gradient. At broad scales, the climatology–anomaly cross component accounts for nearly all of the contraction, whereas the anomaly field's own energy changes little. High regional mean WBT is therefore associated with a flatter relative geographic pattern, rather than uniform smoothing of transient anomalies. The negative raw-field contrast persists under area weighting, alternative kernels, fixed-hour analyses and calendar matching. Its attenuation after site-month and site–year–month demeaning further locates the result in the cross-site climatic gradient.

Several limitations remain. The 126-km graph has the coarsest effective spatial support and the weakest station evidence. The rectangular domain is a fixed sampling support, so other boundaries may produce different allocations. The historical extension uses the same reanalysis product, and the 1950–1978 segment carries the documented preliminary-forcing caveat. The station comparison covers 28 locations in ten summers and uses event times and labels from ERA5-Land; it assesses measurement agreement and station-sampled contrast, but not independent event identification or full-field replication. Pressure-level circulation, population-weighted fields and health outcomes are needed to examine mechanism and exposure. Across the analysed finite record, high-regional-mean days had weaker broad-scale WBT contrast while transient anomaly energy changed little. Generalisation to a long-run climate process requires longer or independent records.

Data availability

ERA5-Land is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land`). The NOAA ISD station archive (Smith et al., 2011) is available at `https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database`. The processed data, site and station manifests, quality-control records, derived results and complete analysis code are available in a private reviewer-access repository. Access details are supplied through ScholarOne. On acceptance, the repository will be made public and assigned a persistent identifier. The raw reanalysis and station archives remain available through their source providers.

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Conflict of interest

The authors declare no conflict of interest.

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Table 1: Data-generating processes in the simulation study. Day-level fields use the observed 121-site coordinates, and all metrics are computed from identical simulated fields and joint shifts.

Experiment	Component	Specification
Day-level	Baseline field	Gaussian anomaly field projected to have exactly zero spatial sample mean; exponential spatial covariance with 450-km range and 5% nugget; temporal coefficient 0.45; six records of 30–31 days.
	Event labels	Regional-mean series, autoregressive with coefficient 0.65 and equal to the simulated field mean; centred anomalies are independent of this series; high and middle labels use within-record quartiles.
	Null variants	One aspect varied at a time: three or 12 records; 20 or 60 days per record; temporal coefficients 0 or 0.75; independent t_3 radial scaling; two non-circular finite autoregressions with coefficients 0.45 and 0.75.
	Alternatives	Anomaly standard deviation $\times 0.97, 0.955, 0.94, 0.92, 0.90$; correlation range 525, 587, 650, 725, 800 km; gradient amplitude 1.6, 1.5, 1.4, 1.3, 1.2 standardised units (baseline 1.8).
	Replication	25 design cells × 1,000 data sets; 499 joint product shifts per test.
Summer-level	Yearly effects	Autoregressive coefficients 0, 0.3, 0.6; marginal standard deviation 0.12; innovations Gaussian, centred log-normal or standardised t_3; 200-year burn-in.
	Sample	$R = 20, 33, 60, 120$ summers; true mean effects $0, -3.5\%, -7\%$; 108 cells × 10,000 replications.

Table 2: Power comparison over all nine spatial alternatives. Entries after the true graph-profile effect are rejection rates (%) from 1,000 simulated data sets. Moran's I and Geary's C use two-sided tests; the five dispersion procedures use lower-tail tests.

Mechanism	Strength	True effect (%)	Graph profile	Five-bin variogram	Variance	Nearest neighbour	Single-bin semivariance	Moran I	Geary C
Amplitude	Weak	−5.91	50.0	33.6	31.1	76.4	45.0	5.2	5.9
	Medium	−11.64	95.3	79.8	77.1	99.8	91.9	4.6	4.5
	Strong	−19.00	99.9	99.3	99.1	100.0	99.9	5.4	4.7
Range	Weak	−6.07	48.9	19.0	16.4	95.4	51.5	15.5	15.9
	Medium	−14.38	98.6	62.4	53.3	100.0	99.1	45.3	53.7
	Strong	−22.20	100.0	91.4	85.6	100.0	100.0	74.0	85.0
Gradient	Weak	−10.34	90.9	93.9	96.5	40.8	78.9	32.4	81.6
	Medium	−19.47	100.0	100.0	100.0	82.5	99.8	87.6	100.0
	Strong	−27.37	100.0	100.0	100.0	97.6	100.0	99.8	100.0

Table 3: Physical scale of the evaluation-period graph contrasts. Q^M and Q^H are equal-record descriptive means in $^\circ\mathrm{C}^2$; RMS columns report $(2Q)^{1/2}$ in °C. The graph effect is the prespecified mean of record-specific squared-difference ratios, so it need not equal the ratio of the two descriptive Q columns. Negative gives the number of summers with a negative graph effect.

Bandwidth (km)	Q^M ($^\circ\mathrm{C}^2$)	Q^H ($^\circ\mathrm{C}^2$)	RMS^M (°C)	RMS^H (°C)	Graph effect (%)	Negative (/33)
126	2.66	2.58	2.31	2.27	−2.86	22
252	4.36	4.17	2.95	2.89	−3.94	23
503	7.94	7.43	3.99	3.86	−5.91	25
1,006	12.89	11.49	5.08	4.79	−10.44	29
2,013	15.90	13.74	5.64	5.24	−13.27	30